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Grade	**100.00** out of 100.00

Question 1

Correct

Mark 25.00 out of 25.00

(The Triangle class) Design a class named **Triangle** that extends **GeometricObject**. The class contains:

- Three double data fields named side1, side2, and side3 with default values 1.0 to denote three sides of a triangle.
- A no-arg constructor that creates a default triangle.
- A constructor that creates a triangle with the specified side1, side2, and side3.
- The accessor methods for all three data fields.
- A method named getArea() that returns the area of this triangle.
- A method named getPerimeter() that returns the perimeter of this triangle.
- A method named toString() that returns a string description for the triangle.

The toString() method is implemented as follows:

return "Triangle: side1 = " + side1 + " side2 = " + side2 + " side3 = " + side3;

You can directly use the **GeometricObject** provided in **Listing 11.1**.

Write a test program that prompts the user to enter three sides of the triangle, a color, and a Boolean value to indicate whether the triangle is filled. The program should create a Triangle object with these sides and set the color and filled properties using the input. The program should display the area, perimeter, color, and true or false to indicate whether it is filled or not.

For example:

Input	Result
5 6 7 blue true	Enter three sides: 5.0 6.0 7.0 Enter the color: blue Enter a boolean value for filled: true The area is 14.696938456699069 The perimeter is 18.0 created on time color: blue and filled: true Triangle: side1 = 5.0 side2 = 6.0 side3 = 7.0

Answer: (penalty regime: 0 %)

Reset answer

```
import java.util.Scanner;

public class Exercise11_01 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter three sides: ");
    double side1 = input.nextDouble();
    double side2 = input.nextDouble();
    double side3 = input.nextDouble();
    System.out.println(side1 + " " + side2 + " " + side3);

    Triangle triangle = new Triangle(side1, side2, side3);

    System.out.print("Enter the color: ");
    String color = input.next();
    triangle.setColor(color);
    System.out.println(color);

    System.out.print("Enter a boolean value for filled: ");
    boolean filled = input.nextBoolean();
    triangle.setFilled(filled);
    System.out.println(filled);

    System.out.println("The area is " + triangle.getArea());
    System.out.println("The perimeter is " + triangle.getPerimeter());
    System.out.println(triangle);
  }
}

class Triangle extends GeometricObject {
  private double side1 = 1.0, side2 = 1.0, side3 = 1.0;


  public Triangle() {
  }


  public Triangle(double side1, double side2, double side3) {
    this.side1 = side1;
    this.side2 = side2;
    this.side3 = side3;
```

```
  }

  public double getSide1() {
    return side1;
  }

  public double getSide2() {
    return side2;
  }

  public double getSide3() {
```

	Input	Expected	Got	
✔	5 6 7 blue true	Enter three sides: 5.0 6.0 7.0 Enter the color: blue Enter a boolean value for filled: true The area is 14.696938456699069 The perimeter is 18.0 created on time color: blue and filled: true Triangle: side1 = 5.0 side2 = 6.0 side3 = 7.0	Enter three sides: 5.0 6.0 7.0 Enter the color: blue Enter a boolean value for filled: true The area is 14.696938456699069 The perimeter is 18.0 created on time color: blue and filled: true Triangle: side1 = 5.0 side2 = 6.0 side3 = 7.0	✔

Passed all tests! ✔

Show/hide question author's solution (Java)

```java
import java.util.Scanner;

public class Exercise11_01 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter three sides: ");
    double side1 = input.nextDouble();
    double side2 = input.nextDouble();
    double side3 = input.nextDouble();
    System.out.println(side1 + " " + side2 + " " + side3);

    Triangle triangle = new Triangle(side1, side2, side3);

    System.out.print("Enter the color: ");
    String color = input.next();
    triangle.setColor(color);
    System.out.println(color);

    System.out.print("Enter a boolean value for filled: ");
    boolean filled = input.nextBoolean();
    triangle.setFilled(filled);
    System.out.println(filled);

    System.out.println("The area is " + triangle.getArea());
    System.out.println("The perimeter is " + triangle.getPerimeter());
    System.out.println(triangle);
  }
}

class Triangle extends GeometricObject {
  private double side1 = 1.0, side2 = 1.0, side3 = 1.0;


  public Triangle() {
  }


  public Triangle(double side1, double side2, double side3) {
    this.side1 = side1;
    this.side2 = side2;
    this.side3 = side3;
  }

  public double getSide1() {
    return side1;
  }

  public double getSide2() {
    return side2;
  }

  public double getSide3() {
    return side3;
  }

```

```
  public double getArea() {
    double s = (side1 + side2 + side3) / 2;
    return Math.sqrt(s * (s - side1) * (s - side2) * (s - side3));
  }


  public double getPerimeter() {
    return side1 + side2 + side3;
  }

  @Override
  public String toString() {
    // Implement it to return the three sides
    return super.toString() +
      "\nTriangle: side1 = " + side1 + " side2 = " + side2 + " side3 = " + side3;
  }
}


class GeometricObject {
 private String color = "white";
 private boolean filled;
 private java.util.Date dateCreated;

 /** Construct a default geometric object */
 public GeometricObject() {
 dateCreated = new java.util.Date();
 }

/** Construct a geometric object with the specified color
 * and filled value */
 public GeometricObject(String color, boolean filled) {
 dateCreated = new java.util.Date();
 this.color = color;
 this.filled = filled;
 }

 /** Return color */
 public String getColor() {
 return color;
 }

 /** Set a new color */
 public void setColor(String color) {
 this.color = color;
 }

 /** Return filled. Since filled is boolean,
 its getter method is named isFilled */
 public boolean isFilled() {
 return filled;
 }

 /** Set a new filled */
 public void setFilled(boolean filled) {
 this.filled = filled;
 }

 /** Get dateCreated */
 public java.util.Date getDateCreated() {
 return dateCreated;
 }

 /** Return a string representation of this object */
 public String toString() {
 return "created on " + "time" + "\ncolor: " + color +
 " and filled: " + filled;
 }
 }
```

Correct

Marks for this submission: 25.00/25.00.

Question 2

Correct

Mark 25.00 out of 25.00

(The Person, Student, Employee, Faculty, and Staff classes)

Design a class named **Person** and its two subclasses named **Student** and **Employee**.

Make Faculty and Staff subclasses of Employee.

- A person has a name, address, phone number, and e-mail address.
- A student has a class status (freshman, sophomore, junior, or senior). Define the status as a constant.
- An employee has an office, salary, and date hired. Use the *java.util.Calendar* to create an object for date hired.
- A faculty member has office hours and a rank. A staff member has a title. Override the toString method in each class to display the class name and the person's name.

Write a test program that creates a Person, Student, Employee, Faculty, and Staff, and invokes their *toString()* methods.

For example:

Input	Result
1	Person: Peter Student: Susan Employee: Eva Faculty: Frank Staff: Shane

Answer: (penalty regime: 0 %)

Reset answer

```
public class Exercise11_02 {
  public static void main(String[] args) {
    Person person = new Person("Peter");
    Student student = new Student("Susan");
    Employee employee = new Employee("Eva");
    Faculty faculty = new Faculty("Frank");
    Staff staff = new Staff("Shane");

    System.out.println(person);
    System.out.println(student);
    System.out.println(employee);
    System.out.println(faculty);
    System.out.println(staff);
  }
}

class Person {
  protected String name;
  protected String address;
  protected String phoneNumber;
  protected String email;

  Person(String name) {
    this.name = name;
  }

  public String getName() {
    return name;
  }

  @Override
  public String toString() {
    return "Person: " + name;
  }
}

class Student extends Person {
  public static final int FRESHMAN = 1;
  public static final int SOPHOMORE = 2;
  public static final int JUNIOR = 3;
  public static final int SENIOR = 4;

  protected int status;

  Student(String name) {
    super(name);
  }

  @Override
```

```
  public String toString() {
    return "Student: " + getName();
  }
```

	Input	Expected	Got	
✔	1	Person: Peter Student: Susan Employee: Eva Faculty: Frank Staff: Shane	Person: Peter Student: Susan Employee: Eva Faculty: Frank Staff: Shane	✔

Passed all tests! ✔

Show/hide question author's solution (Java)

```java
public class Exercise11_02 {
  public static void main(String[] args) {
    Person person = new Person("Peter");
    Student student = new Student("Susan");
    Employee employee = new Employee("Eva");
    Faculty faculty = new Faculty("Frank");
    Staff staff = new Staff("Shane");

    System.out.println(person);
    System.out.println(student);
    System.out.println(employee);
    System.out.println(faculty);
    System.out.println(staff);
  }
}

class Person {
  protected String name;
  protected String address;
  protected String phoneNumber;
  protected String email;

  Person(String name) {
    this.name = name;
  }

  public String getName() {
    return name;
  }

  @Override
  public String toString() {
    return "Person: " + name;
  }
}

class Student extends Person {
  public static final int FRESHMAN = 1;
  public static final int SOPHOMORE = 2;
  public static final int JUNIOR = 3;
  public static final int SENIOR = 4;

  protected int status;

  Student(String name) {
    super(name);
  }

  @Override
  public String toString() {
    return "Student: " + getName();
  }
}

class Employee extends Person {
  protected String office;
  protected int salary;
  protected java.util.Calendar dateHired;

  Employee(String name) {
    super(name);
  }

  @Override
  public String toString() {
    return "Employee: " + getName();
  }
```

```
}

class Faculty extends Employee {
  public static int LECTURER = 1;
  public static int ASSISTANT_PROFESSOR = 2;
  public static int ASSOCIATE_PROFESSOR = 3;
  public static int PROFESSOR = 4;

  protected String officeHours;
  protected int rank;

  Faculty(String name) {
    super(name);
  }

  @Override
  public String toString() {
    return "Faculty: " + name;
  }
}

class Staff extends Employee {
  protected String title;

  Staff(String name) {
    super(name);
  }

  @Override
  public String toString() {
    return "Staff: " + getName();
  }
}

```

Correct

Marks for this submission: 25.00/25.00.

Question 3

Correct

Mark 25.00 out of 25.00

Write a method that returns the union of two array lists of integers using the following header:

public static ArrayList<Integer> union(ArrayList<Integer> list1, ArrayList<Integer> list2)

For example, the addition of two array lists {2, 3, 1, 5} and {3, 4, 6} is {2, 3, 1, 5, 3, 4, 6}.

Write a test program that prompts the user to enter two lists, each with five integers, and displays their union.

The numbers are separated by exactly one space.

For example:

Input	Result
3 5 45 4 3 33 51 5 4 13	Enter five integers for list1: 3 5 45 4 3 Enter five integers for list2: 33 51 5 4 13 The combined list is 3 5 45 4 3 33 51 5 4 13

Answer: (penalty regime: 0 %)

Reset answer

```
import java.util.Scanner;
import java.util.ArrayList;

public class Exercise11_14 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter five integers for list1: ");
    ArrayList<Integer> list1 = new ArrayList<>();
    for (int i = 0; i < 5; i++){
      list1.add(input.nextInt());
      System.out.print(list1.get(i) + " ");
    }

    System.out.println();

    System.out.print("Enter five integers for list2: ");
    ArrayList<Integer> list2 = new ArrayList<>();
    for (int i = 0; i < 5; i++){
      list2.add(input.nextInt());
      System.out.print(list2.get(i) + " ");
    }
    System.out.println();
    ArrayList<Integer> list3 = add(list1, list2);

    System.out.print("The combined list is ");
    for (int i = 0; i < list3.size(); i++)
      System.out.print(list3.get(i) + " ");
  }

  public static ArrayList<Integer> add(ArrayList<Integer> list1, ArrayList<Integer> list2) {
    ArrayList<Integer> result = new ArrayList<>();

    for (int i = 0; i < list1.size(); i++)
      result.add(list1.get(i));

    for (int i = 0; i < list2.size(); i++)
      result.add(list2.get(i));

    return result;
  }
}

```

	Input	Expected	Got	
✔	3 5 45 4 3 33 51 5 4 13	Enter five integers for list1: 3 5 45 4 3 Enter five integers for list2: 33 51 5 4 13 The combined list is 3 5 45 4 3 33 51 5 4 13	Enter five integers for list1: 3 5 45 4 3 Enter five integers for list2: 33 51 5 4 13 The combined list is 3 5 45 4 3 33 51 5 4 13	✔

Passed all tests! ✔

▾ **Show/hide question author's solution (Java)**

```java
import java.util.Scanner;
import java.util.ArrayList;

public class Exercise11_14 {
  public static void main(String[] args) {
    Scanner input = new Scanner(System.in);
    System.out.print("Enter five integers for list1: ");
    ArrayList<Integer> list1 = new ArrayList<>();
    for (int i = 0; i < 5; i++){
      list1.add(input.nextInt());
      System.out.print(list1.get(i) + " ");
    }

    System.out.println();

    System.out.print("Enter five integers for list2: ");
    ArrayList<Integer> list2 = new ArrayList<>();
    for (int i = 0; i < 5; i++){
      list2.add(input.nextInt());
      System.out.print(list2.get(i) + " ");
    }
    System.out.println();
    ArrayList<Integer> list3 = add(list1, list2);

    System.out.print("The combined list is ");
    for (int i = 0; i < list3.size(); i++)
      System.out.print(list3.get(i) + " ");
  }

  public static ArrayList<Integer> add(ArrayList<Integer> list1, ArrayList<Integer> list2) {
    ArrayList<Integer> result = new ArrayList<>();

    for (int i = 0; i < list1.size(); i++)
      result.add(list1.get(i));

    for (int i = 0; i < list2.size(); i++)
      result.add(list2.get(i));

    return result;
  }
}

```

Correct

Marks for this submission: 25.00/25.00.

Question 4

Correct

Mark 25.00 out of 25.00

(Remove duplicates) Write a method that removes the duplicate elements from an array list of integers using the following header:

public static void removeDuplicate(ArrayList<Integer> list)

Write a test program that prompts the user to enter 10 integers to a list and displays the distinct integers in their input order and separated by exactly one space.

For example:

Input	Result
34 5 3 5 6 4 33 2 2 4	Enter ten integers: 34 5 3 5 6 4 33 2 2 4 The distinct integers are 34 5 3 6 4 33 2

Answer: (penalty regime: 0 %)

Reset answer

```
import java.util.Scanner;
import java.util.ArrayList;

public class Exercise11_13 {
  public static void main(String[] args) {
    System.out.print("Enter ten integers: ");
    ArrayList<Integer> list = new ArrayList<>();
    Scanner input = new Scanner(System.in);
    for (int i = 0; i < 10; i++) {
      list.add(input.nextInt());
      System.out.print(list.get(i) + " ");
    }
    System.out.println();
    removeDuplicate(list);

    System.out.print("The distinct integers are ");
    for (int i = 0; i < list.size(); i++)
      System.out.print(list.get(i) + " ");
  }

  public static void removeDuplicate(ArrayList<Integer> list) {
    ArrayList<Integer> temp = new ArrayList<>();
    for (int i = 0; i < list.size(); i++)
      if (!temp.contains(list.get(i)))
        temp.add(list.get(i));

    list.clear();
    for (int i = 0; i < temp.size(); i++)
      list.add(temp.get(i));
  }
}
```

	Input	Expected	Got	
✔	34 5 3 5 6 4 33 2 2 4	Enter ten integers: 34 5 3 5 6 4 33 2 2 4 The distinct integers are 34 5 3 6 4 33 2	Enter ten integers: 34 5 3 5 6 4 33 2 2 4 The distinct integers are 34 5 3 6 4 33 2	✔

Passed all tests! ✔

Show/hide question author's solution (Java)

```
import java.util.Scanner;
import java.util.ArrayList;

public class Exercise11_13 {
  public static void main(String[] args) {
    System.out.print("Enter ten integers: ");
    ArrayList<Integer> list = new ArrayList<>();
    Scanner input = new Scanner(System.in);
    for (int i = 0; i < 10; i++) {
      list.add(input.nextInt());
      System.out.print(list.get(i) + " ");
    }
    System.out.println();
    removeDuplicate(list);

    System.out.print("The distinct integers are ");
```

```
    for (int i = 0; i < list.size(); i++)
      System.out.print(list.get(i) + " ");
  }

  public static void removeDuplicate(ArrayList<Integer> list) {
    ArrayList<Integer> temp = new ArrayList<>();
    for (int i = 0; i < list.size(); i++)
      if (!temp.contains(list.get(i)))
        temp.add(list.get(i));

    list.clear();
    for (int i = 0; i < temp.size(); i++)
      list.add(temp.get(i));
  }
}
```

Correct

Marks for this submission: 25.00/25.00.